Sol-n

(b) Using the superposition prin. show by integrating wrt λ, we get

$$\frac{x}{x^2+y^2}, \quad \frac{y}{x^2+y^2} \quad \text{bounded har. fns.}$$

Note that by the result of part (a), the only bounded harmonic f-ns are

(1) $$F.S.S(x) = \{ e^{-\lambda y} \cos(\lambda x), \quad e^{-\lambda y} \sin(\lambda x) \}, \quad \lambda \geq 0.$$

So by superposition principle the following

(2) $$U_1(x,y) = \lim_{n\to\infty} \sum_{\lambda=0}^{\infty} e^{-\lambda y} \cos(\lambda x) = \int_0^\infty e^{-\lambda y} \cos(\lambda x)\, d\lambda$$

(3) $$U_2(x,y) = \lim_{n\to\infty} \sum_{\lambda=0}^{\infty} e^{-\lambda y} \sin(\lambda x) = \int_0^\infty e^{-\lambda y} \sin(\lambda x)\, d\lambda$$

U_1 & U_2 must be $\in C^0$ (upperhalf plane $y>0$), bounded (integral of bounded f-ns) & of course satisfy the Laplace eq-n and hence harmonic.

So (2) is

(5) $$U_1 = \int_0^\infty e^{-\lambda y} \cos(\lambda x)\, d\lambda =$$

$$= \frac{x e^{-\lambda y} \cdot \sin(\lambda x) - y \cdot e^{-\lambda y} \cdot \cos(\lambda x)}{x^2 - y^2} \Bigg|_{\lambda=0}^{\infty} = \frac{-y}{x^2-y^2}$$

(details are on the white board page.)

$$U_1 = \int e^{-\lambda y} \cos(\lambda x)\, d\lambda = \int \underbrace{e^{-\lambda y}}_{u}\, d\underbrace{\left(\frac{\sin(\lambda x)}{x}\right)}_{v} = e^{-\lambda y} \cdot \frac{\sin(\lambda x)}{x} - \int \frac{\sin(\lambda x)}{x} (-y) \cdot e^{-\lambda y}\, d\lambda = e^{-\lambda y} \cdot \frac{\sin(\lambda x)}{x} + \frac{y}{x} \int e^{-\lambda y} \overbrace{\sin(\lambda x)\, d\lambda}^{\int \cdot\, d\lambda} \Rightarrow$$

$$d\left[\frac{\cos(\lambda x)}{-x}\right]$$

$$U_1 = e^{-\lambda y} \cdot \frac{\sin(\lambda x)}{x} + \frac{y}{x} \left\{ \underbrace{e^{-\lambda y}}_{u} \cdot d \underbrace{\left[\frac{\cos(\lambda x)}{-x}\right]}_{v} \right\} = e^{-\lambda y} \cdot \frac{\sin(\lambda x)}{x} + \frac{y}{x} \left\{ e^{-\lambda y} \cdot \left[\frac{\cos(\lambda x)}{-x}\right] - \int \frac{\cos(\lambda x)}{-x} \cdot (-y) \cdot e^{-\lambda y} \cdot d\lambda \right\}$$

$$U_1 = \frac{e^{-\lambda y} \sin(\lambda x)}{x} - \frac{y}{x^2} \cdot e^{-\lambda y} \cdot \cos(\lambda x) + \frac{y^2}{x^2} \cdot \underbrace{\int e^{-\lambda y} \cos(\lambda x)\, d\lambda}_{U_1}$$

$$\left[1 - \frac{y^2}{x^2}\right] U_1 = \frac{x e^{-\lambda y} \sin(\lambda x) - y \cdot e^{-\lambda y} \cos(\lambda x)}{x^2} \; ; \Big\} * \; x^2 \text{ of BS.}$$

$$x^2 \cdot \frac{x^2 - y^2}{x^2} \cdot U_1 = \frac{x e^{-\lambda y} \sin(\lambda x) - y \cdot e^{-\lambda y} \cos(\lambda x)}{x^2} \cdot x^2$$

(5) $$\boxed{U_1 = \frac{x e^{-\lambda y} \sin(\lambda x) - y \cdot e^{-\lambda y} \cdot \cos(\lambda x)}{x^2 - y^2}}$$

is a harmonic bounded f-n on our defined domain Ω.

Similarly,

$$U_2 = \int_0^{+\infty} e^{-\lambda y} \sin(\lambda x)\, d\lambda = \int_0^{\infty} \underbrace{e^{-\lambda y}}_{u} \cdot d\underbrace{\left[\frac{\cos(\lambda x)}{-x}\right]}_{v} =$$

$$= e^{-\lambda y} \cdot \frac{\cos(\lambda x)}{-x}\Bigg|_0^{\infty} - \int_0^{\infty} \frac{\cos(\lambda x)}{-x} \cdot (-y) \cdot e^{-\lambda y}\, d\lambda =$$

$$= \frac{e^{-\lambda y} \cdot \cos(\lambda x)}{-x} - \frac{y}{x} \int e^{-\lambda y} \cos(\lambda x)\, d\lambda \quad \text{so} \Rightarrow$$

$$U_2 = \frac{e^{-\lambda y} \cdot \cos(\lambda x)}{-x} - \frac{y}{x} \left\{ \int e^{-\lambda y} \cdot d\left[\frac{\sin(\lambda x)}{x}\right] \right\} =$$

$$= \frac{e^{-\lambda y} \cdot \cos(\lambda x)}{-x} - \frac{y}{x} \left\{ \frac{e^{-\lambda y} \cdot \sin(\lambda x)}{x} - \int \frac{\sin(\lambda x)}{x} \cdot (-y) \cdot e^{-\lambda y}\, d\lambda \right\}$$

$$= \frac{e^{-\lambda y} \cdot \cos(\lambda x)}{-x} - \frac{y}{x^2} \cdot e^{-\lambda y} \cdot \sin(\lambda x) + \frac{y^2}{x^2} \cdot \underbrace{\int e^{-\lambda y} \sin(\lambda x)\, d\lambda}.$$

$$U_2 = \frac{e^{-\lambda y} \cdot \cos(\lambda x)}{-x} - \frac{y}{x^2} e^{-\lambda y} \sin(\lambda x) + \frac{y^2}{x^2} \cdot U_2 \Rightarrow$$

$$\left[1 - \frac{y^2}{x^2}\right] \cdot U_2 = -\left[\frac{x e^{-\lambda y} \cos(\lambda x) + y e^{-\lambda y} \sin(\lambda x)}{x^2}\right]$$

$$\times x^2 \Big) \Rightarrow \bar{U}_2\left[\frac{y^2}{x^2}-1\right] = \left[\frac{x^{-\lambda y}\cos(\lambda x) + y\, e^{-\lambda y}\sin(\lambda x)}{x^2}\right]$$

$$\bar{U}_2 \frac{y^2-x^2}{x^2}\cdot x^2 = \frac{x^{-\lambda y}\cos(\lambda x) + y\, e^{-\lambda y}\sin(\lambda x)}{x^2}\cdot x^2$$

(6) $$\boxed{\bar{U}_2 = \frac{x^{-\lambda y}\cos(\lambda x) + y\, e^{-\lambda y}\sin(\lambda x)}{y^2-x^2}\Bigg|_0^\infty = \frac{-x}{-(y^2-x^2)}}$$

So, the F.S.S is

(7) $$F.S.S = \left\{\frac{x}{x^2-y^2}, \frac{y}{x^2-y^2}\right\}$$

I do not know for some reason by double checking I'm still getting x^2-y^2 in the denominator.